QUANTUM MACHINE LEARNING · QML-2026 FDP · SESSION 1

clustering, reimagined

quantum unsupervised learning — finding structure in unlabelled data with quantum kernels

Archit Srivastava · E&ICT Academy NIT Warangal × NIT Raipur · 11 July 2026

ABOUT THE SPEAKER

Archit Srivastava

5+ research papers	25+ invited talks
10+ universities	200+ faculty engaged

quantum × ai · photonic hardware · gw & black holes · enterprise tech

- Senior Technical Consultant, **o9 Solutions** — enterprise AI & decision systems
- Founder, **AiQyaM** — India's first quantum-hardware community, est. **2020**
- Founder, **CIRQuIT Quantum Research**, RV College of Engineering
- Senior Research Associate, **Quantum Computing India**
- **GKQCTP & IQDC** with Innogress Ventures — toward a sovereign Quantum Tech Park
- Quantum PoCs at **HPE** & **o9** · early intern **BosonQ Psi** · **Conf42** speaker
- Published: **Quantum Finance** (VQE, Qiskit vs PennyLane) & **AI computer vision** with a quantum-visual-tracking outlook

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CURRENT FOCUS

THE PATH TODAY

First principles, bottom-up.

1 · The idea

What "unsupervised" means, and why clustering is how we make first sense of any new dataset.

2 · The classical tools

k-means, its blind spot, and the spectral method that fixes it — the bridge to quantum.

3 · The quantum turn

Qubits, the swap test, and the quantum kernel — a new kind of similarity.

4 · The honest picture

A live demo, where the advantage is real, and what it means for India.

WHY NOW · 2025 WAS THE YEAR QUANTUM GOT REAL

From promise to measured advantage.

Three peer-reviewed firsts in a single year moved quantum advantage from a talking point to a number you can check.

Google Willow

0

- superconducting qubits, **Nature** Dec 2024
- first **below-threshold** error correction — logical error shrinks as the code grows
- benchmark in <5 min vs ~10^{25} yr classically

D-Wave Advantage2

0+

- qubit annealer, **Science** Mar 2025
- magnetic-materials simulation in minutes
- a **real scientific target** — parts later contested on GPUs

Quantum Echoes

0×

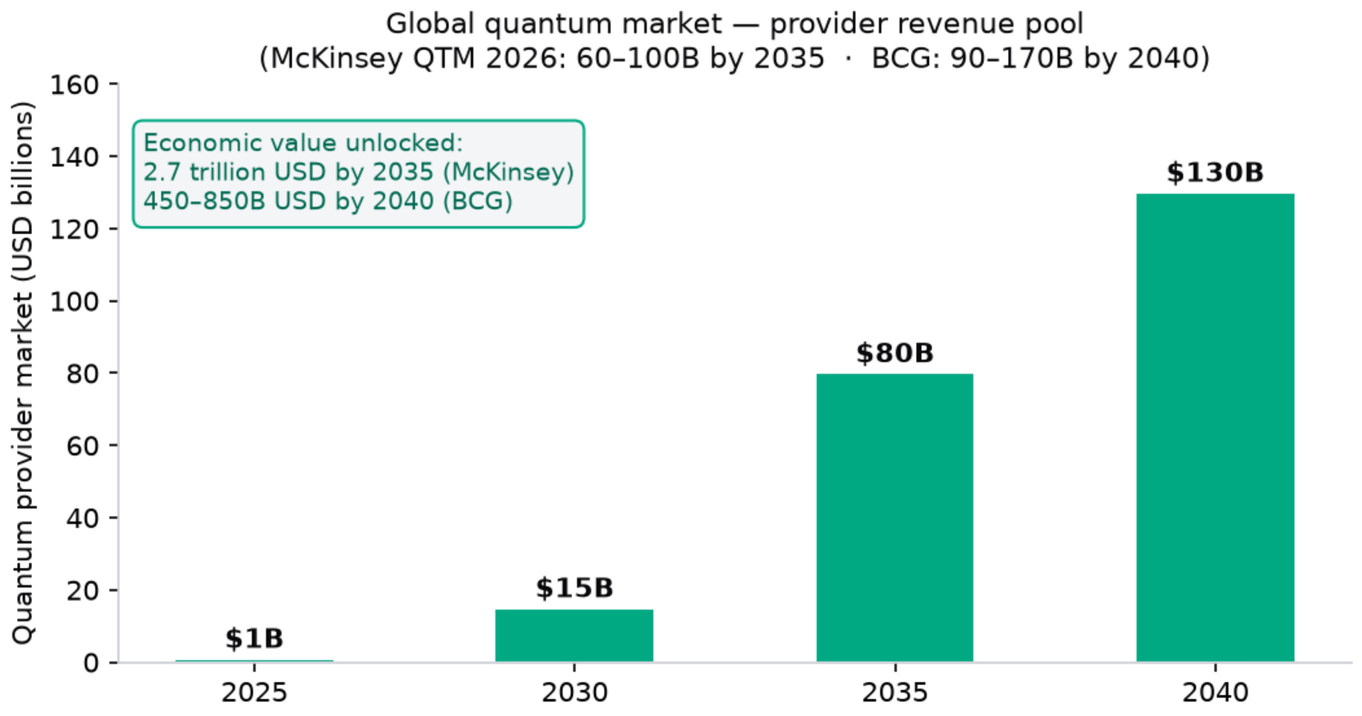
- Google, **Nature** Oct 2025
- first **verifiable** advantage — checkable on a second machine
- a path toward molecular-structure readouts

The shift that matters for this room: the field has crossed from "can a quantum computer do anything hard?" to "on which problems, and can we verify it?" That is the question your students will spend their careers answering.

WHY NOW · THE ECONOMICS

The value isn't in selling processors — it's in the outcomes.

GLOBAL MARKET · MCKINSEY QTM 2026 · BCG



THE SIGNAL

- **$2.7 trillion** economic value unlocked by 2035 (McKinsey)
- internal market **$60–100B by 2035**; QC alone $43–71B
- start-up investment **$12.6B in 2025 — 6.3× YoY**, ~90% to quantum computing
- industry needs **~840,000 workers by 2035**; ~5,000 qualified in 2025
- the value is captured by **software, algorithms & AI** — a keyboard-and-brains problem

$2.7T
economic value by 2035 (McKinsey)

6.3×
YoY jump in quantum investment, 2025

~840k
quantum jobs needed by 2035 — the talent gap

WHY NOW · THE SOVEREIGN BET

India's National Quantum Mission

A national bet across mission funding, thematic hubs and state valleys — building hardware, secure networks and talent so India isn't dependent on a foreign quantum stack.

INDIA

The mission

₹6,003 cr

- approved **2023** · 8-year horizon to **2031**
- staged target **50–1,000 physical qubits**
- QKD to **2,000 km**; sensors, atomic clocks, materials
- ~half the targets reportedly met within **3 years**

Four thematic hubs · 43 institutions

- **Computing** — IISc Bengaluru
- **Communication** — IIT Madras + C-DOT
- **Sensing** — IIT Bombay
- **Materials** — IIT Delhi
- 152 researchers · 17 states — a **national footprint**

Shipping, not slideware

0 qubits

- **QpiAI** "Kaveri" 64-qubit (Nov 2025); "Indus" 25-qubit
- **IISER Pune** ion-trap, 20 qubits due end-2026
- **Amaravati Quantum Valley** — IBM System Two, launched Feb 2026
- ~**1,000 km** quantum comms demonstrated · national **PQC roadmap**

THE IDEA · FIRST PRINCIPLES

Sorting coins in the dark.

Unsupervised learning finds structure in data that has **no labels**. You're handed a pile of points and asked: which of these belong together?

Tip a bag of mixed coins onto a table in the dark. You can't read the denominations — but you can feel **size** and **weight**. You sort them into piles by similarity.

That sorting — no labels, only a notion of "alike" — is clustering.

THE ML MAP

- **Supervised:** data + labels → learn a mapping. "10,000 labelled tumours → predict the next."
- **Unsupervised:** data only → discover structure. "10,000 tumours — are there natural subtypes?"
- Labels are expensive, scarce, or impossible. **~80% of real data is unlabelled.**
- Uses: customer segments, genomic subtypes, anomaly detection, image compression, document topics.

THE IDEA · THE RECIPE

Every clustering method is three choices.

1 · A representation

Each object becomes a vector of features — a point in $\mathbb{R}^d$. A face, a transaction, a gene, a pixel patch.

2 · A similarity

How we measure "alike" — Euclidean distance, cosine similarity, or a **kernel**. This is the ingredient quantum changes.

3 · An objective

What makes a grouping good — minimise within-cluster spread, maximise between-cluster separation.

Hold this thought — it's the spine of the whole session: **quantum methods mostly change ingredient 2, the similarity, and how fast we can compute it.**

THE CLASSICAL TOOLS · THE WORKHORSE

k-means: nearest-centroid reasoning.

LLOYD'S ALGORITHM

- **Initialise** k centroids (k-means++).
- **Assign:** each point → nearest centroid. This is n×k distance computations — remember this line.
- **Update:** each centroid → mean of its points.
- Repeat until assignments stop changing.

$$J = \sum_k \sum_{x \in C_k} \|x - \mu_k\|^2$$

STRENGTHS & THE BLIND SPOT

- simple, fast, scales to millions of points
- must pick **k** in advance
- assumes round, equal-size, linearly-separable blobs

The blind spot: "nearest centroid" is a straight-line idea. On interleaved crescents or concentric rings, the nearest centroid is often in the wrong cluster. k-means fails on non-convex shapes.

THE CLASSICAL TOOLS · THE LANDSCAPE

Beyond k-means — and the bridge to quantum.

Hierarchical

Build a tree of merges (dendrogram). No need to fix k. $O(n^2–n^3)$.

DBSCAN

Density-based. Finds arbitrary shapes, labels outliers. Needs a density scale ε.

GMM

Soft, probabilistic clusters via Expectation-Maximisation.

Spectral ★

Similarity graph → top eigenvectors of its Laplacian → k-means there. **Handles non-convex clusters.**

Spectral clustering is the natural bridge to quantum: its heavy step is linear algebra on a similarity / kernel matrix — and a quantum computer is a native linear-algebra machine. Hold that; we return to it.

THE CLASSICAL TOOLS · WHERE THE COST LIVES

The recurring bill: distances and similarity matrices.

$O(d)$

one Euclidean distance in d dimensions

$O(n \cdot k \cdot d)$

one k-means iteration

$O(n^2 d) \rightarrow O(n^3)$

build a kernel matrix → eigendecompose it

Everything expensive in clustering is **computing similarities between many high-dimensional vectors** and **linear algebra on n×n similarity matrices**. That is exactly the corner where quantum offers leverage — inner-product estimation and high-dimensional feature spaces. Now we earn the quantum part.

THE QUANTUM TURN · THE RESOURCE

A qubit is a coin that is every angle at once.

ONE QUBIT

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle\ ,\ |\alpha|^2 + |\beta|^2 = 1$

- α, β are complex **amplitudes**; $|\alpha|^2$ = probability of measuring 0
- **Superposition:** before measurement it genuinely holds both
- **Bloch sphere:** $|0\rangle$ = north pole, $|1\rangle$ = south — a qubit is a continuous direction, not one bit

N QUBITS → 2^n AMPLITUDES

$2^{50} \approx 10^{15}$

50 qubits carry **a quadrillion amplitudes** evolving together. That exponential state space is the resource.

The catch (be honest): measurement collapses the state and returns only n classical bits. You can't read out all 2^n amplitudes. The art is arranging **interference** so the useful answer is what you measure.

Load a vector into amplitudes; read out an overlap.

Amplitude encoding

Load a classical vector **x** into the amplitudes of a state |x⟩. A d-dimensional vector needs only **⌈log₂ d⌉ qubits** — exponential compression.

$$|x\rangle = (1/\|x\|)\ \Sigma_i\ x_i|i\rangle$$

Overlap = similarity

The natural quantity between two states is their **overlap** ⟨a|b⟩. Inner products are similarities. This is the whole game.

$$\|a-b\|^2 = 2 - 2\langle a|b\rangle$$

Interference + measure

A small circuit turns an overlap into a **measurable probability**. That circuit is the swap test — next slide.

THE QUANTUM TURN · THE HERO PRIMITIVE

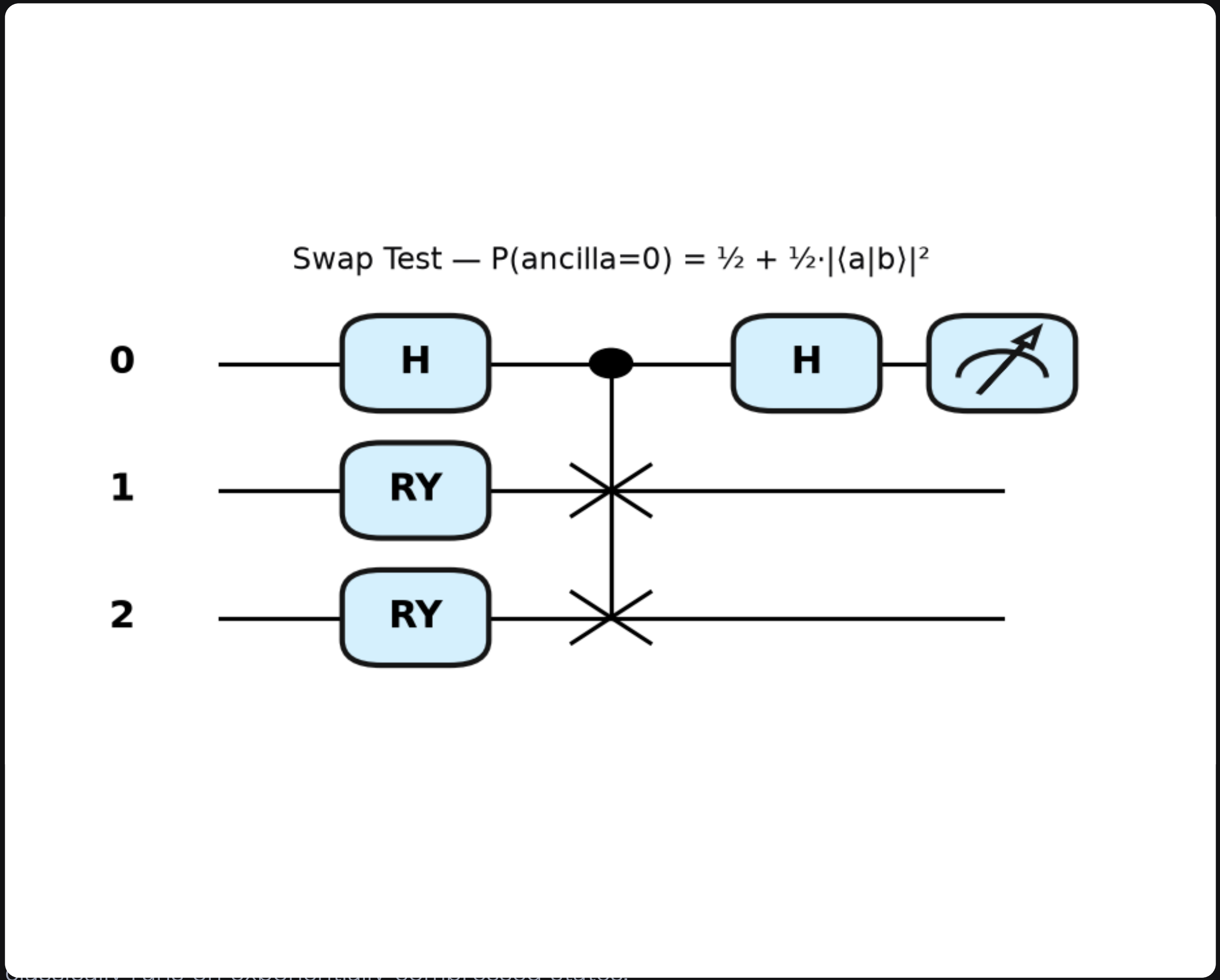
The Swap Test — similarity in one measurement.

THE CIRCUIT, IN WORDS

- Prepare $|a\rangle$, $|b\rangle$; ancilla in $|0\rangle$
- **Hadamard** the ancilla → superposition
- **Controlled-SWAP:** ancilla conditionally swaps the registers
- **Hadamard** the ancilla again; measure it

$$P(\text{ancilla} = 0) = \tfrac{1}{2} + \tfrac{1}{2}\cdot|\langle a|b\rangle|^2$$

Similar → P_0 near 1. Orthogonal → $P_0 = \tfrac{1}{2}$. Run many shots, invert, and you have the similarity — of two **d-dimensional** states with a **constant-size** circuit.



THE QUANTUM TURN · THE NISQ-FRIENDLY ROUTE

The quantum kernel — a similarity hard to fake classically.

THE IDEA

Don't speed up k-means — **change the similarity** to one that is expensive classically. A **quantum feature map** φ encodes each point into a quantum state via a parameterised circuit. The kernel is the overlap:

$$K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$$

- Feed K into any kernel method — **spectral clustering, kernel k-means**
- The bet: the feature map reaches a **Hilbert space expensive to simulate**, so hidden structure becomes separable
- **Shallow & NISQ-friendly** — runs on today's hardware

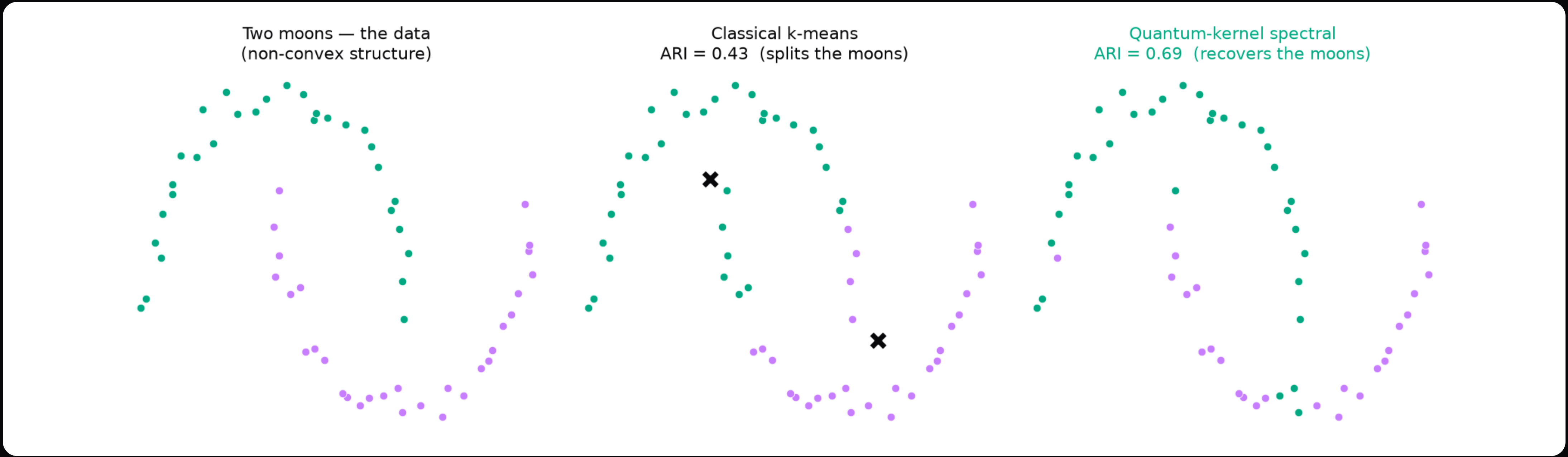
THIS CLOSES THE LOOP

Spectral clustering (the classical method that handles non-convex data) + a quantum kernel (a richer similarity) = **the near-term, runnable approach.**

It's exactly what the notebook demonstrates next.

LIVE DEMO · QUANTUM-KERNEL CLUSTERING

k-means fails on the moons. The quantum kernel doesn't.



0.43
k-means ARI — splits the interleaved moons

0.69
quantum-kernel spectral ARI — recovers them

2 qubits
data re-uploading feature map · runs on a laptop

FROM MY OWN WORK · VARIATIONAL OPTIMIZATION

The same circuits, put to work on portfolios.

QUANTUM FINANCE — AN OVERVIEW (EASYCHAIR 6071, 2021)

- Mapped **mean-variance portfolio optimization** → QUBO → Ising Hamiltonian, solved with a **VQE**
- Implemented in **both Qiskit and PennyLane** — the framework choice this room faces
- Qiskit: optimal selection **[0 1 0 1]** at probability **0.94**; PennyLane VQE converged to **[1 0 1]**

The **VQE ansatz** (RX/RY rotations + CNOT entanglers) is the same species of circuit as today's **quantum feature map**.

There we optimised it to minimise a Hamiltonian. Here we evaluate it to measure similarity. Same physics, two jobs.

THE HONEST PICTURE

What is real, and what waits on hardware.

THE CAVEATS — SAY THEM OUT LOUD

- **Data loading is the bottleneck.** The exponential k-means speedup assumes **QRAM**, which doesn't yet exist at scale.
- **Dequantization (Tang, 2018+):** several "exponential" QML speedups were later matched classically under the same assumptions.
- **NISQ noise.** Devices are shallow and noisy — which is why shallow quantum kernels are attractive.
- **Barren plateaus:** deep variational maps can have vanishing gradients — keep circuits shallow/structured.

WHERE THE ADVANTAGE IS GENUINE

- quantum-native / structured data
- small-but-hard kernels a classical machine can't simulate
- as a **feature generator** feeding the classical methods you already teach

Not "k-means but faster on your CSV." A new kind of similarity you can run now.

THE CLOSE · WHY THIS ROOM MATTERS

Quantum is a pillar of Viksit Bharat 2047.

A developed India by the centenary of independence — and the frontier technologies that get us there.

THE ARGUMENT

- India already took frontier tech to **population scale** — Aadhaar, UPI, Digital India. That capability is proven.
- The next leap: **"world's back office"** → **"world's innovation engine."** Frugal jugaad is necessary but no longer sufficient.
- A coordinated mission stack: **NQM + IndiaAI + Supercomputing + Semicon + ANRF's ₹1 lakh crore RDI fund.**
- India's edge is the **algorithm & application layer** — exactly where QML, clustering and quantum vision live.

"Becoming a leader in quantum will prove central to Viksit Bharat — India can lead the era of quantum algorithm & application discovery."

— IBM, on India's path to Viksit Bharat 2047

2–3%	2047
target R&D / GDP by 2047	centenary of independence

THE CLOSE · YOUR MOVE

The bottleneck is people. You are the lever.

Teach it

Clustering, kernels and the swap test are teachable from first principles — no hardware required. Seed one module, one project, one reading group. The global shortage of people who understand this stack is India's opening.

Build with it

The notebooks from today run on a laptop — PennyLane, scikit-learn, no hardware needed. Point a student at them this week.

Connect to the mission

NQM hubs, state quantum valleys and startups (QpiAI, QNu Labs) need talent. Your classroom is the top of that funnel.

Viksit Bharat 2047 will be built by the cohort sitting in your classrooms right now. Every course you seed is India's comparative advantage compounding.

thank you

Archit Srivastava · QML-2026 FDP · NIT Warangal × NIT Raipur

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